

Neurofeedback Training of the Control Network in Children Improves Brain Computer Interface Performance

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Abstract—In the past 20 years, neural engineering has made unprecedented progress in the interpretation of brain information (e.g., brain-computer interfaces) and in neuromodulation (e.g., electromagnetic stimulation and neurofeedback). However, there has been little research aiming to improve the performance of brain-computer interfaces (BCIs) using neuromodulation. The present study presents a novel design for a neurofeedback training (NFT) method to improve the operation of a steady-state visual evoked potential (SSVEP)-based BCI and further explores its underlying mechanisms. The use of NFT to upregulate alpha-band power in the user's parietal lobe is presented in this study as a new neuromodulation method to improve SSVEP-based BCI in this study. After users completed this NFT intervention, the signal-to-noise ratio (SNR), accuracy, and information transfer rate (ITR) of the SSVEP-based BCI were increased by 5.8%, 4.7%, and 15.6%, respectively. However, no improvement was observed in the control group in which the subjects did not participate in NFT. Moreover, a general reinforcement of the information flow from the parietal lobe to the occipital lobe was observed. Evidence from a network analysis and an attention test further indicates that NFT improves attention by developing the control capacity of the parietal lobe and then enhances the above SSVEP indicators. Upregulating the amplitude of parietal alpha oscillations using NFT significantly improves the SSVEP-based BCI performance by modulating the control network. The study validates an effective neuromodulation method and possibly contributes to explaining the function of the parietal lobe in the control network. © 2021 Published by Elsevier Ltd on behalf of IBRO.

Key words: alpha oscillation, neurofeedback training, attention, steady-state visual evoked potential, control network.

INTRODUCTION

The study of cortical oscillations through electroencephalography (EEG) has made significant progress. The effects and influences of various rhythms on people have been gradually explored since neural oscillations were first discovered by Berger (1929). However, the physiological mechanisms by which cortical oscillations affect human behavior are still not fully understood. Some studies have indicated that the roles of neural oscillations include feature binding, information transfer mechanisms, and the generation of rhythmic motion output (Klimesch, 1999; Buzsaki, 2004; Canolty et al., 2006; Buzsáki and Moser, 2013; Iaccarino, 2016).

In recent decades, a few breakthroughs have been made in this field through electroencephalography and neuroimaging. It has been found that neural oscillations play an essential role in neural information processing. However, there is little experimental evidence regarding the function of neural oscillations thus far. It is unlikely that the role of neural oscillation can be explained based on the existing evidence alone. Further insights into neural oscillations and their relationship to cognitive processes are necessary. Furthermore, attempts should be made to regulate and control the specific characteristics of these oscillations to achieve purposeful functional changes.

Alpha-band oscillations are the dominant oscillations in the human brain. Although the physiological basis of these oscillations is still unclear, their origin is thought to be related to the thalamo-cortical circuitry (Lopes da Silva et al., 1973; Nunez et al., 2001; Nicolelis and Faselow, 2002; Lőrincz et al., 2009; Hughes et al., 2011; Schmid et al., 2012; Howells et al., 2018). An increasing body of recent research on alpha-band function has shown that the alpha band performs a significant

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Abbreviations: BCI, brain-computer interface; EEG, electroencephalography; ERD, event-related desynchronization; FBCCA, filter bank canonical correlation analysis; ITR, information transfer rate; NFT, neurofeedback training; SNR, signal-to-noise ratio; SSVEP, steady-state visual evoked potential; TRCA, task-related component analysis.

inhibitory function in information processing by the brain. Previous work has suggested that information is routed by functionally blocking off task-irrelevant pathways (gating by inhibition). Importantly, this inhibition is reflected by oscillatory activity in the alpha band. The task dependent activation and inhibition of different regions form part of the mechanism of attention (Jensen and Mazaheri, 2010). Moreover, evidence has shown that alpha-band oscillations reflect cognitive and memory performance and state. Alpha-band oscillations enable voluntary orientation of attention and controlled knowledge access, since they are closely linked to the fundamental processes of attention regulation (suppression and selection) (Klimesch, 2012). The traditional argument regarding the role of alpha-band activity in attention is that alpha oscillations desynchronize in response to anticipatory attention and in the absence of stimuli. This type of event-related desynchronization (ERD) is termed a “pre-stimulus, anticipatory ERD”. A large prestimulus ERD is relevant to high detection performance (Ergenoglu et al., 2004; Hanslmayr et al., 2005a; Hanslmayr et al., 2005b) and reflects the maintenance of target information in an attentional buffer. Furthermore, an increase in the amplitude of event-related synchronization (ERS) reflects inhibition, whereas a decrease in the amplitude of ERS reflects release from inhibition. Examples have been found in tasks that vary the side on which stimulation is applied (e.g., stimulation of the right vs. left visual hemifield), the stimulus modality, or the stimulus processing domain (e.g., related to the ventral vs. dorsal processing stream). Findings show that when attention is focused on the auditory element of a compound auditory-visual stimulus, alpha oscillation is increased in the visual cortices (Foxe et al., 1998). When a task engages the ventral stream, alpha power is increased in parietal regions (Jokisch, 2007). Moreover, the ability of humans to respond to visual stimuli based on their spatial locations requires the parietal cortex (Corbetta, 2000) to direct visual attention. One theory maintains that the parietal cortex controls the autonomous orientation of attention toward a location of interest (Mesulam, 1981; Heilman et al., 1985; Halligan and Marshall, 1994). Another emphasizes its role in reorienting attention toward visual targets appearing at unattended locations (Posner et al., 1984; Morrow and Ratcliff, 1988; Di Pellegrino, 1995; Friedrich et al., 1998). All of the abovementioned studies suggest that the parietal lobe is closely related to attention to visual stimuli and plays a crucial role in control and monitoring.

Neurofeedback training (NFT) is an operant conditioning method that helps subjects to control or change their brain activity (Schafer and Moore, 2011). The training enables the voluntary modulation of neural oscillations through real-time feedback on the brain's electrical parameters. Neurofeedback training is regarded as a safer and more stable method than electromagnetic stimulation for controlling or changing neural oscillations. Since its inception, NFT has been applied in several scenarios (Zoefel et al., 2011), such as efforts to cure attention-deficit hyperactivity disorder (Arns et al., 2009; Sonuga-Barke et al., 2013), epilepsy (Zhao et al., 2009),

and Huntington's disease (Papoutsis et al., 2017); to improve cognitive performance (Vernon et al., 2003; Hanslmayr et al., 2005a; Hanslmayr et al., 2005b); and to enhance self-regulation of emotional and behavioral control (Mayeli et al., 2019).

At present, NFT is also applied to improve the operation of various types of BCIs. Steady-state visual evoked potential (SSVEP)-based interfaces are a classical BCI paradigm that forms an essential part of many BCIs for various applications. SSVEP can be understood as phase-locked human brain activity that spontaneously responds to stimulation; the signal-to-noise ratio (SNR) is the measure most commonly used to evaluate the signal quality. Despite decades of research, the performance of SSVEP-based BCIs is still heavily relies on the specific subjects and algorithms (Neuper and Pfurtscheller, 2010). SSVEP signals and performance can be influenced by NFT to such an extent that Mayra, in a recent study on an SSVEP-based BCI algorithm, excluded subjects who had undergone any form of NFT before data collection (Mayra and Maurits, 2018). A study by Ordikhani-Seyedlar suggested that the power of the SSVEP signal was regulated by various degrees of attention, whether covert or overt (Ordikhani-Seyedlar et al., 2014). Additionally, Collura and Thomas designed experiments to demonstrate the relationship between periodic SSVEPs and short-term attention (Collura and Thomas, 2002). Moreover, their work pointed out that attention-sensitive SSVEPs in response to stimulation could serve as the basis for a type of neurofeedback and could likely be used in training to overcome attention deficits. Regarding the application of neurofeedback derived from SSVEPs, a study by Bruno has claimed that visual and auditory cognition can be improved by binocular audio and SSVEP neurofeedback (Bruno et al., 2017).

Morgan provided further direct evidence regarding the relationship between attention and SSVEPs. Specifically, he asked subjects to gaze at the center of a screen and showed blocks of flicker stimuli of different frequencies on both the left and right sides (Morgan et al., 1996). This attention-bias experiment found that subjects' EEG signals reflected the frequency of stimulation on the side to which attention was being focused, thus demonstrating the correlation between attention and SSVEPs.

Based on the above descriptions, since alpha oscillation exerts an inhibitory physiological function and SSVEP-based BCI performance appears to require considerable attention, we hypothesized that improving the alpha amplitude in the parietal lobe can increase attention and the SNR of the SSVEP signal to enhance SSVEP performance.

EXPERIMENTAL PROCEDURES

Design

The flow chart in Fig. 1 shows the experimental design in detail. For each subject in the NFT group, the experiment consisted of twenty-four training sessions within two months. Participants were assigned to complete three sessions per week, with each session lasting an hour. In

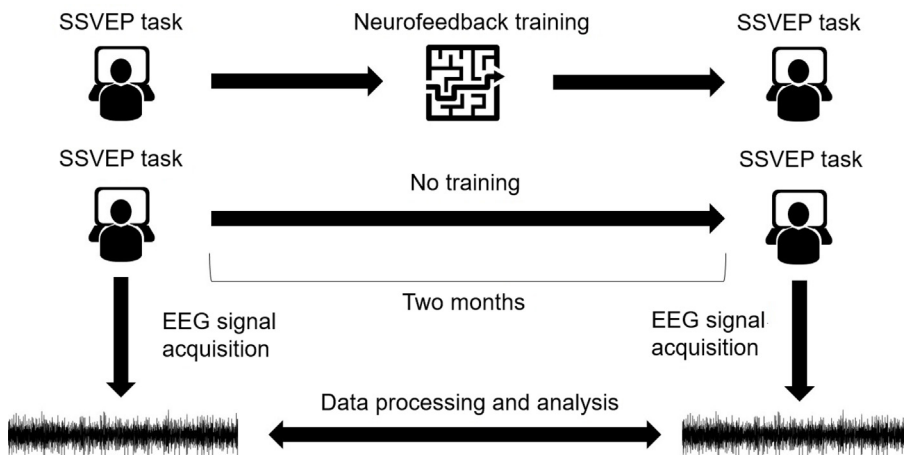


Fig. 1. Description of protocol, from pretraining to posttraining, for the NFT group and non-NFT group. The full experimental process included data acquisition, feedback training, and data processing. The control group and the experimental group differed only in whether feedback training was performed.

each session, subjects were asked to play an EEG feedback game. Each week, they were free to perform their three training sessions at any time as long as the sessions were separated by at least one day. To capture the subjects' interest, the feedback interface was presented to the subjects in a gamified manner with a controllable process. The SSVEP typing test was arranged to compare performance approximately ten days before and after the two months of training.

Participants

A total of forty-seven subjects, comprising twenty-five in the NFT group and twenty-two in the control group (randomly assigned), took part in the experiment. Subjects with normal or corrected vision and no epilepsy or other diseases were selected. All participants' guardians signed a written consent form. Moreover, the project was reviewed by the Ethics Committee of the Tsinghua University School of Medicine. The final sample consisted of 24 subjects in the NFT group (six females and eighteen males, aged 9.42 ± 2.58 years) and 22 subjects in the control group (eight females and fourteen males, aged 9.86 ± 3.14 years). As van Praag's experiment suggested, aging causes changes in the hippocampus that may lead to cognitive and neural plasticity decline in older adults (van Praag, 2005). Accordingly, this research chose children to participate in NFT because young brains are more neuroplastic than older brains.

EEG recordings

EEG data were recorded from 64 electrodes positioned according to the international 10–20 system (AF3/4, AF7/8, Fp1/2, FPz, Fz, F1/2, F3/4, F5/6, F7/8, FC1/2, FC3/4, FC5/6, FCz, FT7/8, Cz, C1/2, C3/4, C5/6, T3/4, T7/8, TP7/8, CP1/2, CP3/4, CP5/6, Pz, P3/4, P5/6, P7/8, PO3/4, PO5/6, PO7/8, POz, Oz, O1/2, HEOL, HEOR, ECG, VEOL, VEOU) and referenced to CPz. A

portable wireless EEG amplifier (NeuSen.W64, Neuracle, China) was used for data recording at a sampling rate of 1000 Hz. Electrode impedances were kept below 10 k Ω for all electrodes throughout the experiment. The experiment was carried out in an unmodified office environment without any electrical shielding.

Alpha-band power ratio

In this study, NFT was performed to increase the alpha-band amplitude. The alpha-band power was calculated to examine the training outcome. Since the absolute value of the alpha-band amplitude is affected by the current state of the subject and the environment (Eqilimi et al., 2020; Schubring and Schupp, 2021), values are not com-

parable between different trials. To contrast the NFT results, we calculate the alpha-band power ratio P_{α} to characterize whether the alpha amplitude was improved. In the following formula, X_1 and X_2 denote the power of the alpha band and the full spectrum, respectively. In the formula, X denotes the EEG signal and f denotes the frequency.

$$F(\omega) = \text{fft}(X)$$

$$f_1 = \sum_8^{12} |F(\omega)|^2$$

$$f_2 = \sum_{0.8}^{20} |F(\omega)|^2$$

$$P_{\alpha} = \bar{f}_1 / \bar{f}_2 \quad (1)$$

The data used to calculate the alpha-band power ratio were obtained from two recordings, one with the eyes open and the other with the eyes closed, lasting 2 minutes each. The closed-eye data totaled 120,000 sampling points with a sampling rate of 1000, and the frequency-domain resolution was 1/12 Hz. Thus, the step size for summation was 1/12 Hz, and the same value was used in formulas 3 and 4.

SSVEP stimulation

The processing system consists of a stimulation device, an EEG acquisition device, and a processing computer. The stimulation device was an ASUS MG279Q 27-inch display with 2560*1440 resolution and a refresh rate of 143 Hz. The processing device is an Intel i9-9900 K@3.6 GHz CPU and DDR4 32 GB memory. Stimulation and processing programs were developed using MATLAB 2015a and Psychophysics Toolbox Version 3.

During the experiment, the stimulation was presented in the form of a multitarget flicker. A start trigger and an end trigger signal were sent to the EEG collector at the

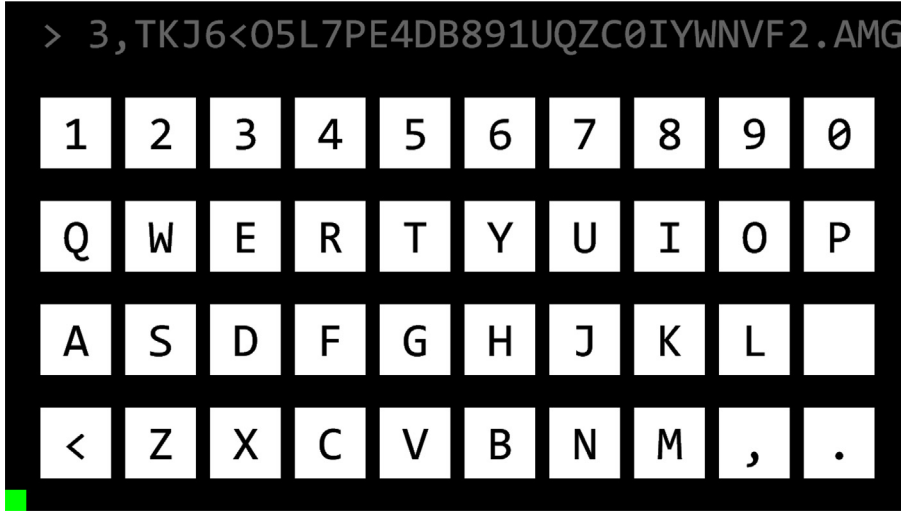


Fig. 2. The SSVEP stimulation interface consisted of 40 stimulus target blocks on the screen, which represented the letters A–Z (a total of 26 letters), the digits 0–9 (a total of ten digits), a space, a comma, a period, and a backspace key.

beginning and end of each trial. The processing device received the EEG data from the EEG amplifier in real time and returned the processed results to the stimulation device after the identification was completed.

As Figs. 2 and 3 show, in the SSVEP test, a 40-target SSVEP speller based on frequency phase modulation coding (frequency range: 8–15.8 Hz, frequency interval: 0.2 Hz) was used (Chen et al., 2015a; Chen et al., 2015b). A total of 40 stimulus target blocks were displayed on the screen; these targets represented the letters A–Z (a total of 26 letters), the digits 0–9 (a total of ten digits), a space, a comma, a period, and a backspace key. In the experiment, each stimulus target was constructed by sampling the sinusoidal method (Manyakov et al., 2013; Chen et al., 2014).

In the entire SSVEP task, to smooth out fluctuations in the actual state of subjects and ensure the validity of the data, the staff required each subject to complete six blocks (40 trials per block). Participants could choose to rest between blocks to maximize their performance. Thus, the entire test took approximately 40 minutes.

The entire online test (with feedback results in real time) was divided into six blocks. The first four blocks used a 3- to 4-second dynamic window and were classified by applying the filter bank canonical correlation analysis (FBCCA) algorithm (Chen et al., 2015a; Chen et al., 2015b). The last two blocks used a 1- to 2-second dynamic window and applied the task-related component analysis (TRCA) algorithm for classification (Nakanishi et al., 2017), where the training data for TRCA were the four preceding blocks. For each target, data from 6 trials were available to calculate the result.

The average SNR of nine leads in the occipital lobe (PO3/4, PO5/6, PO7/8, Oz, O1/2) and the information transfer rate (ITR) were examined to estimate the

SSVEP performance of the subjects. In the formulas below, N is the total target number (40 in total), B represents the block number (6 in total), A represents accuracy, and T represents the time required to identify. In formula 6, the summation step size is 1.

In this study, the SNR was used as an indicator of SSVEP performance to minimize the effect of background EEG activity across subjects. Therefore, the power of the signal is specified as the power of a narrow frequency interval centered on the stimulus frequency ($f \pm 0.2$). The power of a broader scale ($f \pm 1$) is defined as noise. The response spectrum suggested that these parameters could separate adjacent frequencies and accurately characterize the SNR. In the

formula, $F(\omega)$ denotes the EEG signal in the frequency domain.

$$\text{ITR} = [\log_2 N + A * \log_2(A) + (1 - A) * \log_2(1 - A/N - 1)] * 60/T \quad (2)$$

$$P_{\text{signal}} = \sum_{f_{\text{target}} - 0.2}^{f_{\text{target}} + 0.2} |F(\omega)|^2 \quad (3)$$

$$P_{\text{noise}} = \sum_{f_{\text{target}} - 1}^{f_{\text{target}} + 1} |F(\omega)|^2 - P_{\text{signal}} \quad (4)$$

$$\text{SNR}_{\text{single_trial}} = 10 * \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \quad (5)$$

$$\text{SNR}_{\text{project}} = (\sum_{b=1}^B \sum_{n=1}^N \text{SNR}_{\text{single_trial}}) / (B * N) \quad (6)$$

Brain information flow analysis

The whole brain can be represented as a complex network where a neuron or a group of nerve cells is regarded as a node, while a neural connection between such nodes is regarded as an edge. Human response to stimulation and emotional behavior might represent the handling of information by a network. The directed transfer function (DTF) is a traditional network analysis method. the formula is as follows, with X representing the EEG signal:

$$X = [x_1(t), x_2(t), \dots, x_n(t)]^T \quad (7)$$

Then, a multivariate autoregressive model (MVAR) was used to fit X. In the formula, A is a model parameter, and E is white noise, while q represents fitting order.

$$\sum_{k=0}^q A(k)X(t-k) = E(t) \quad (8)$$

8Hz 0π	8.8Hz 0π	9.6Hz 0π	10.4Hz 0π	11.2Hz 0π	12Hz 0π	12.8Hz 0π	13.6Hz 0π	14.4Hz 0π	15.2Hz 0π
8.2Hz 0.5π	9Hz 0.5π	9.8Hz 0.5π	10.6Hz 0.5π	11.4Hz 0.5π	12.2Hz 0.5π	13Hz 0.5π	13.8Hz 0.5π	14.6Hz 0.5π	15.4Hz 0.5π
8.4Hz 1π	9.2Hz 1π	10Hz 1π	10.8Hz 1π	11.6Hz 1π	12.4Hz 1π	13.2Hz 1π	14Hz 1π	14.8Hz 1π	15.6Hz 1π
8.6Hz 1.5π	9.4Hz 1.5π	10.2Hz 1.5π	11Hz 1.5π	11.8Hz 1.5π	12.6Hz 1.5π	13.4Hz 1.5π	14.2Hz 1.5π	15Hz 1.5π	15.8Hz 1.5π

Fig. 3. SSVEP frequency and phase table; the frequencies and phases of the stimulation were set according to the paradigm presented in the figure.

Formula 8 was converted to the frequency domain, with $H(f)$ defined as the transfer matrix of the system. In the calculation of the brain DTF, $H(f)$ represents the signal connection strength between two electrodes.

$$X(f) = A^{-1}(f)E(f) = H(f)E(f) \quad (9)$$

$$A(f) = \sum_0^p A(k)e^{-j2\pi f\Delta tk} \quad (10)$$

The connection matrix $H(f)$ contains the information transfer characteristics of the whole network. According to the definition, formula 11 is the DTF.

$$\theta^2(f) = |H_{ij}(f)|^2 \quad (11)$$

When DTF is redefined as y_{ij}^2 , the formula to calculate information flow for channel m is presented below, with i and j representing electrodes, while n represents the total number of electrodes:

$$flow_{in} = \sum_{j=1}^n y_{mj}^2 \quad (12)$$

$$flow_{out} = \sum_{i=1}^n y_{im}^2 \quad (13)$$

In this study, the DTF and information flow were computed with the eConnectome Toolbox (Babiloni et al., 2005; Wilke et al., 2010; He et al., 2011) developed by He Bin's team. With reference to Yan's work (Yan and Gao, 2011), the information flow gain ρ was also calculated in this study. The information flow gain ρ is defined as the ratio of the outflow information to the inflow information (Yan and Gao, 2011). The higher the ρ value of the node, the more significant the contribution of the node to the transmission of network information. The smaller the value of ρ is, the lower the contribution of the node to the transfer of network information.

$$\rho = \frac{flow_{out}}{flow_{in}} \quad (14)$$

Feedback procedure

To determine the feedback position, this study analyzed the global brain information connection network induced by SSVEP using the information flow method. In Fig. 4, the ρ values of the nodes are sorted from largest to smallest. The central nodes (FC1, FCz, FC2, C1, Cz, C2, CP1, CP2) play an essential role in the induction network. The higher the ρ value of the node, the more significant the contribution of the node in the transmission of network information. Consistent with this, Corbetta, using event-related functional magnetic resonance (ER-fMRI), showed that distinct parietal regions mediated different attentional visual target processes (Corbetta, 2000). Thus, the parietal lobe plays a central role in visual tasks that involve reorienting attention. The electrode with the lowest alpha amplitude in the parietal lobe was selected as the source of feedback for NFT. The selected electrode varied from subject to subject. The training targeted only one electrode position.

The NFT system was arranged as shown in Fig. 5. Considering that children have stronger neuroplasticity than adults, subjects aged 7–12 years were selected for this study. To attract the subjects' interest and encourage them to cooperate, the interactive neurofeedback interface was designed in the form of a game (the interface display is shown in Fig. 5). The program gave the subjects three options: a. "Pacman" (in which a character eats beans along a given route), b. "Fly the airplane" (in which an aircraft flies along a given route), and c. "Racing car" (in which the player

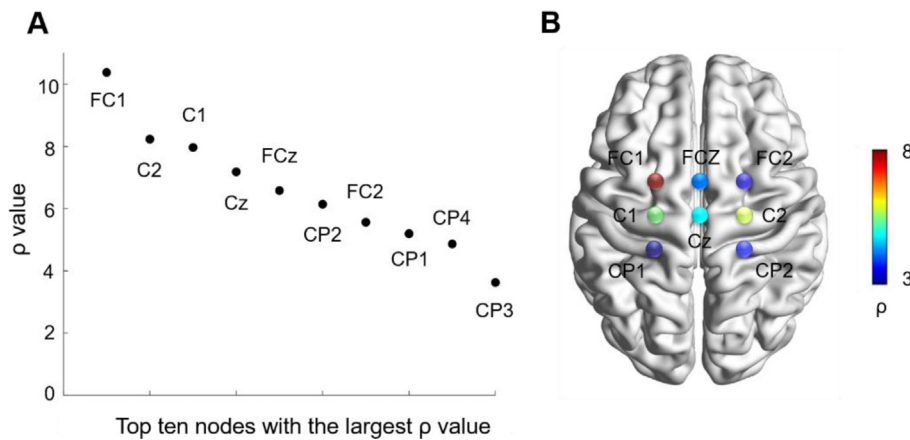


Fig. 4. The functional connectivity nodes that are important in the SSVEP task. In Panel A, the p values of the nodes are sorted from largest to smallest. Panel B shows the locations of the core nodes of the parietal lobe on the cortex.

drives a car on a given route). In all three options, gameplay was paused if the feedback parameters failed to meet the set criteria, and the game progressed when the parameters were in the acceptable range. Regardless of the form of the game, it was always feasible for subjects to keep the EEG parameters above the feedback parameters, demonstrating the effect of NFT.

The setting of feedback parameters was divided into two parts: (1) frequency setting and (2) amplitude setting. Before formal training, the subjects were tested for different feedback frequencies in the alpha band. The frequency at which the success rate of NFT equaled 60%–70% was set as the feedback frequency. On an empirical basis, the amplitude was set to 0.2–0.3 times the alpha amplitude recorded when subjects had

their eyes closed. Once again, the success rate of the training was 60%–70%. The control of the feedback process is explained in Fig. 6. In addition, we adjusted the feedback parameters appropriately according to the success rate (usually increasing the feedback frequency) as the training proceeded. The data were resampled at 250 Hz during calculation, as opposed to the original sampling rate of 1000 Hz. The length of each calculation window was 2 seconds, and the calculation windows overlapped by 50%. The data update time was not to exceed 1 second.

A total of 2000 points were used to compute the feedback parameters. The data were resampled at 250 Hz during calculation, whereas the original sampling rate was 1000 Hz. The length of each calculation window was 2 seconds, and the calculation windows overlapped by 50%. The update time was not to exceed 1 second. Welch's method was used for power computation, and a bandpass filter (0.5–40 Hz) was applied to the original data.

Subjects involved in NFT were instructed to sit in front of the computer and avoid excessive movement. They gazed at the screen and followed the instructions to complete the task while wearing an EEG cap on their head. Participants remained calm and relaxed during the training. The system provided feedback on EEG parameters in real time and determined whether the

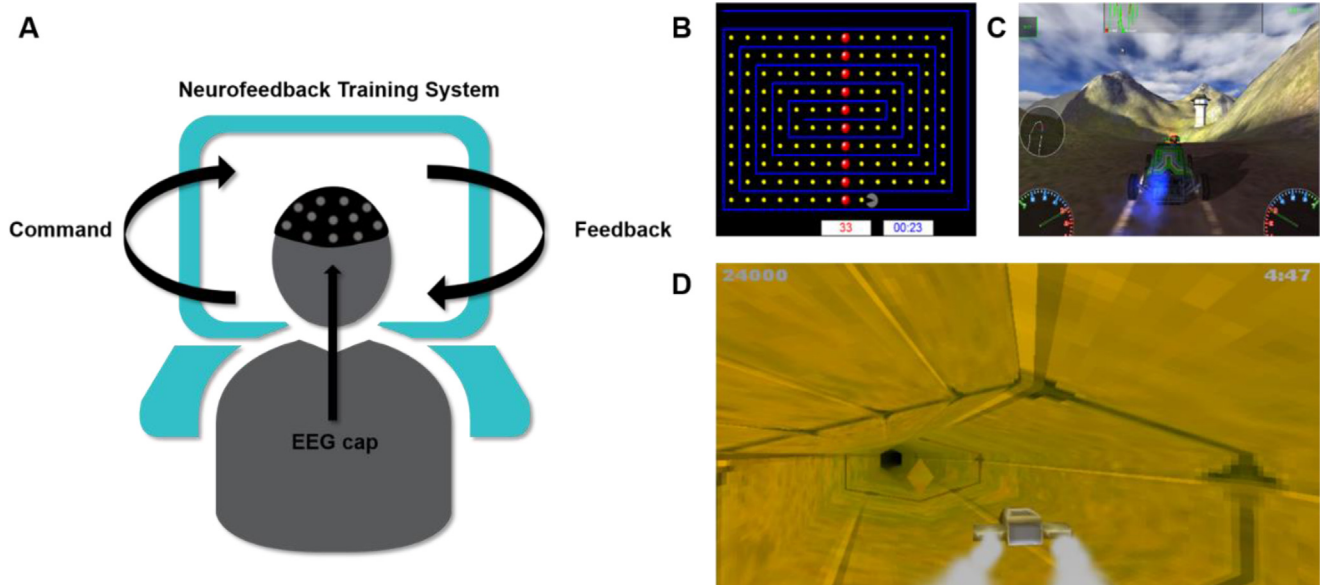


Fig. 5. The feedback system and interface. (A) The subject wears an EEG cap and views the feedback stimulation program, while the computer receives the subject's EEG data in real time and calculates their feedback parameters (frequency and amplitude). The calculated feedback parameters control the feedback procedure, and the effect is returned to the subject. Repeated loops of this process constitute a complete NFT system. Participants choose among three game interfaces designed to excite their interest: (B) "Pacman". (C) "Racing car". (D) "Fly the airplane".

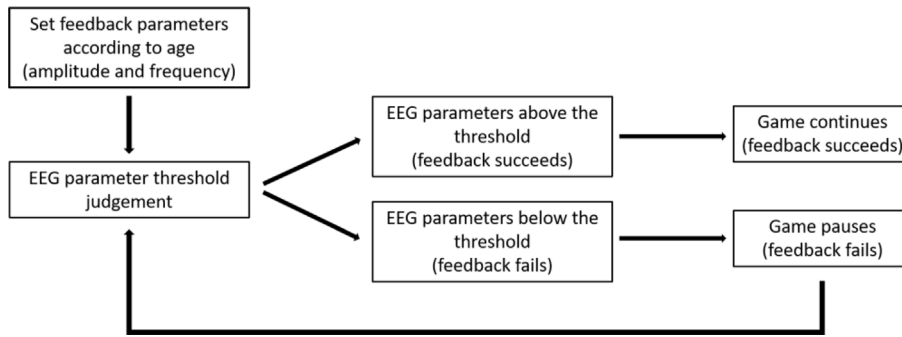


Fig. 6. The NFT process is shown in the figure. The subject's real-time EEG parameters were collected and compared with the set threshold. If all EEG parameters were above the threshold, the game continued. Otherwise, the game was suspended, and the subject needed to adjust his or her state as required. The entire process repeated cyclically to achieve the training goal.

respondents are successful. In the case of success (feedback parameters above the threshold), subjects could continue the game. Otherwise, the staff instructed the subjects to adjust their status and directed their attention toward the game; the subjects could use any strategy to achieve the training goal.

Integrated visual and auditory continuous performance test

The integrated visual and auditory (IVA) continuous performance test (CPT) is a screening tool testing sustained attention (Sandford and Turner, 2000). It was developed by John Sandford (a psychologist) and Anne Turner (a physician). The initial test consisted of a series of pseudorandomized auditory and visual stimuli (500 trials in total); subjects were asked to click on the target they heard or saw. The entire test lasted approximately 20 minutes, and we used our research to select a subset of parameters with which to examine subjects. The subjects completed the test on paper under the guidance of the instructor. The test was a real-time task consisting of repeated auditory and visual stimulations. The instructor guided the subjects to observe a series of graphics and text on the scale. Subjects also listened to a reading of a short essay presenting overt information. The instructor then asked the participants to provide the information required and answer the questions on the given scale. The test is generally used to monitor the effectiveness of NFT or medication.

RESULTS

NFT results

We recruited twenty-four participants; each one completed 24 training sessions in two months (for a total training duration of 24 h). Among these participants, 91.7% were successfully trained (22 of 24). Successful training meant that a subject's alpha-band power was higher after training than before. As shown in Fig. 7, the average alpha-band power ratios of the 24 subjects before and after training were 32.93 and 38.58, respectively. Thus, the training improved the alpha-band power ratio by approximately 17.16%, and a paired

sample *t*-test indicated that the difference was significant ($t = 23$, $p = 0.00015$). Here and throughout, the 't' in the brackets represents the number of degrees of freedom.

Attention test results

Based on the conclusions of numerous existing studies (Klimesch, 2012), the alpha band is closely related to attention, and increasing the alpha amplitude can improve attention. In this study, we used the IVA-CPT, which is the classical test of attention control, to detect the effect of training on

attention. A total of 24 subjects from the experimental group were tested before and after the training, and the test data of 6- to 15-year-old children in Kim's experiment were used as a reference (Kim et al., 2015). Kim's study tested 100 healthy children with the same standard behavioral tests, and we mapped their scores (the original score range was 0–155) to a scale of 0–10 for comparison. The statistical results, shown in the following table, were scored from 0–10, with 10 being the best. As Table 1 shows, the mean value of the 7 factors before training was 5.33 ± 0.97 , while the mean value after training was 6.76 ± 0.93 ; a paired-sample *t*-test ($t = 23$, $p = 0.00172$) confirmed that the difference was significant. This finding indicates that the training brought about marked improvements in visual and auditory attention.

SSVEP results

Improving SSVEP performance is one of the ultimate goals of this work, and we anticipated that such improvement could be observed after NFT. On statistical analysis, the results showed that 87.5% of the subjects (21 of 24) had an improvement.

Accuracy results

Each participant took part in an SSVEP test of 240 trials within 6 blocks. The average accuracy across these 6 blocks was calculated as BCI accuracy (Fig. 8). The results showed that 79.17% of the subjects (19 of 24) improved in some way, while the other five subjects did not improve, as they achieved a relatively high accuracy rate (approximately 90%) on the first test. The average accuracy before training was 78.9%, and the accuracy after training was 83.6%; this difference was statistically significant ($t = 23$, $p = 0.0000049$). As for the control group, the average accuracy before training was 79%, and the accuracy after training was not significantly different, at 79.4%.

ITR results

According to the formula, the average ITR of SSVEPs in the 24 participants was 103. After training, the average score in the same 24 subjects was 119, representing an

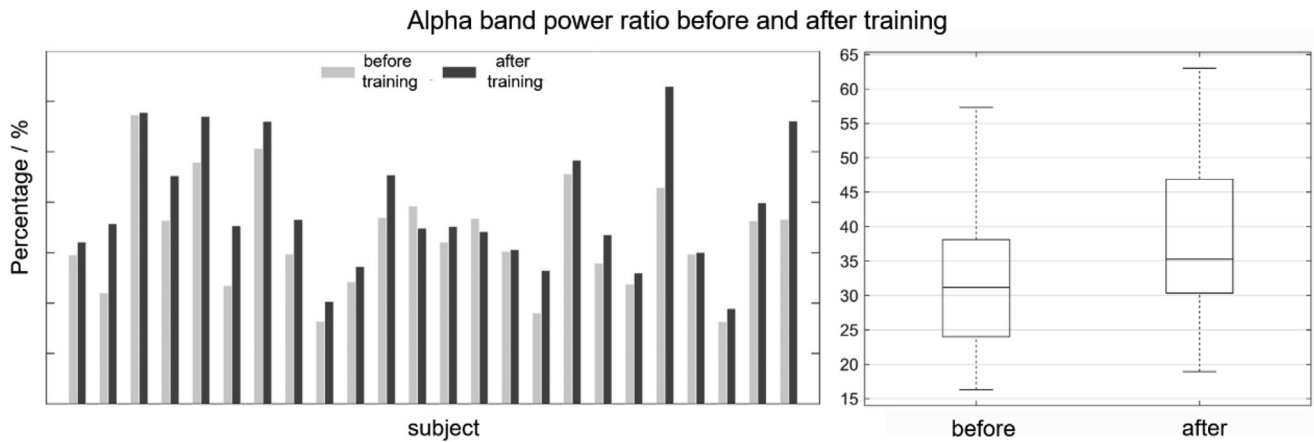


Fig. 7. Neurofeedback training successfully improved the alpha-band power ratio in 22 of 24 subjects. The figure shows the training results of the 24 subjects in the experimental group, of whom 22 subjects showed various degrees of improvement, while two subjects' alpha-band power did not change substantially. In the aggregate, these results suggest that NFT was successful. The statistically averaged alpha-band power ratio and variance are shown in the right-hand figure, and a statistically significant difference can be seen between the results before and after training. On each box, the central mark indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively. The whiskers extend to the highest and lowest data points that are not considered outliers, and any outliers would be plotted individually using the '+' symbol.

Table 1. Results of attention quality tests

Factor	Before training	After training	<i>p</i>	Reference (Kim et al., 2015)
Stability of auditory attention	5.65 ± 0.93	6.82 ± 0.98	0.017	6.97 ± 1.08
Sensitivity of auditory attention	5.81 ± 0.91	6.35 ± 1.12	0.00034	6.64 ± 1.15
Concentration of visual attention	5.53 ± 0.85	6.84 ± 0.91	0.000071	7.19 ± 1.01
Stability of visual attention	5.05 ± 1.23	6.67 ± 0.97	0.00037	7.15 ± 0.81
Distinction of visual attention	4.23 ± 0.95	6.42 ± 0.89	0.0000013	6.92 ± 0.81
Tracking of visual attention	5.09 ± 1.12	6.75 ± 1.31	0.0023	6.93 ± 0.96
Dynamic quality of attention	5.95 ± 0.82	7.45 ± 0.96	0.000026	7.09 ± 0.93
Average score	5.33 ± 0.97	6.76 ± 0.93	0.0029	6.98 ± 0.96

improvement of approximately 15.6% (see Fig. 8). A paired-sample *t*-test indicated that the difference was significant ($t = 23$, $p = 0.0494$). For the control group, the average ITR after two months was only 103, compared to 112 before.

Notably, three of the subjects failed to display an increase in SSVEP ITR because they achieved a high level on the first test, then showed a posttraining decline in this performance index within a normal fluctuation range.

SNR results

The SNR is used in this study as a measure of EEG signal quality. Overall, 79.17% of the participants (19 of 24) were trained to improve their SSVEP SNR. The average SNR of the subjects was -15.2 dB, and the average SNR of the same subjects after training was significantly different, at -14.7 dB ($t = 23$, $p = 0.0116$). For the control group, the average SNR before training was -15.3 , and there was no significant change after two months during the same task (see Fig. 8). The NFT improved the SNR of the experimental group by 0.50 dB. From the perspective of SNR, the training improved the SSVEP signal by 5.8%. Furthermore, the

average SNR changes in 8 frequency bands (8–9 Hz, 9–10 Hz, 10–11 Hz, 11–12 Hz, 12–13 Hz, 13–14 Hz, 14–15 Hz, 15–16 Hz) were calculated (see Fig. 8). Taken together, the evidence indicates that the SNR increased in every frequency band.

We also analyzed the SSVEP frequency responses and used the responses at 8, 9, 10, 11, 12, 13, 14, and 15 Hz for all trials to calculate topographic maps (see Fig. 8). As shown by the SSVEP frequency response topographic maps, the experimental group and the control group responded similarly in the first test. However, in the second test, the experimental group had less interference and a stronger signal than the control group.

We performed a paired *t*-test on the SSVEP frequency response before and after the two-month training period. The resulting *p*-values were plotted as a statistical topographic map, where the scales ranged from 0.2 to 0. Darker colors indicated smaller *p*-values and larger differences; lighter colors indicated larger *p*-values. The locations with significant differences were AF3, FC4, CP1, Pz, and O2, the last of which showed an especially pronounced difference in the experimental group. These results show that the training brought about differences in the parietal, occipital, and frontal

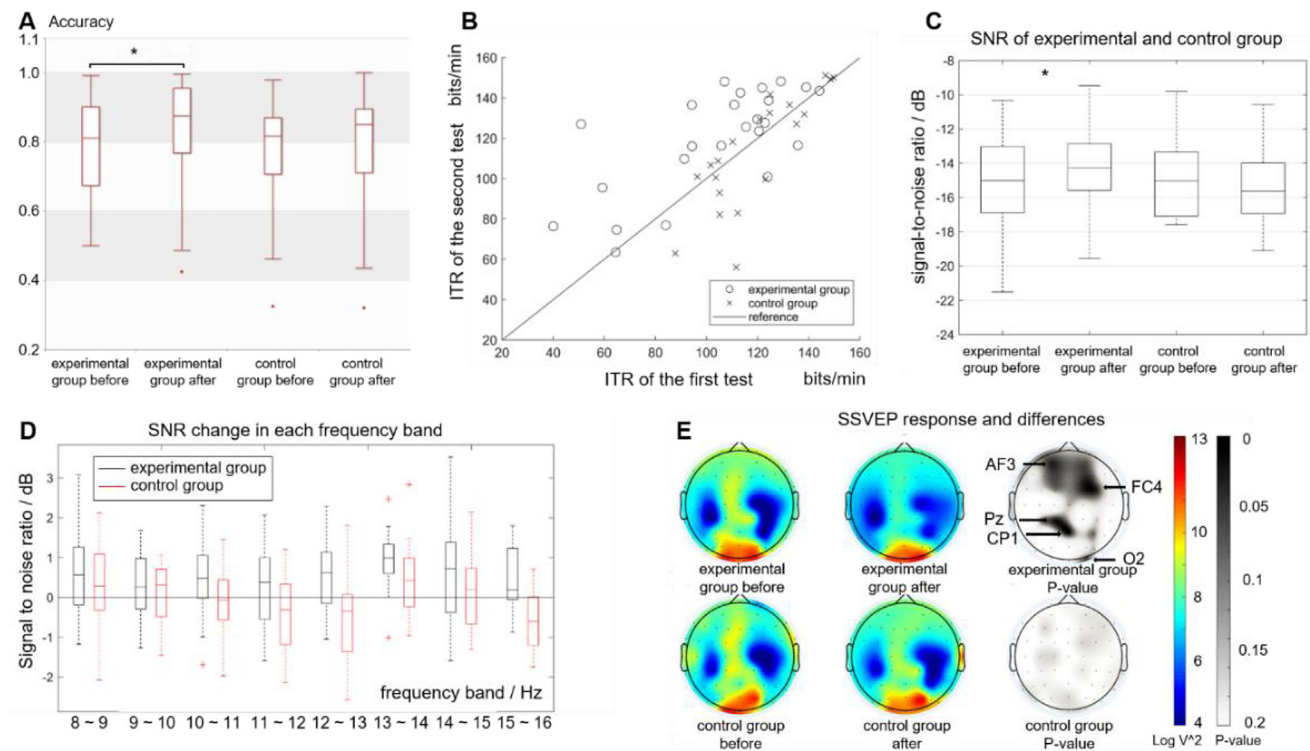


Fig. 8. BCI results. **(A)** The accuracy of SSVEP-based BCI operation is dramatically increased after NFT. The figure shows the distribution of accuracy achieved with the SSVEP-based BCI accuracy. The box plot of the statistical recognition accuracy distribution makes it clear that training dramatically improves the classification accuracy. The results before and after training were significantly different in the experimental group, but there was almost no difference in the control group. It can also be seen from the diagram that the distribution of the initial test results of the experimental group is approximately the same as that of the control group; all individual outliers have been marked. **(B)** The NFT speeds up the ITR of the SSVEP system (50% chance of resampling to avoid singularity, repeated for 100 iterations). The circles indicate the results of the experimental group, and the crosses indicate the ITR of the control group. The diagonal line is the reference line. The figure shows that the experimental group tends to be distributed in the upper left, indicating that the posttraining results are better than the pretraining results. The control group tended to remain at the same performance level, near the diagonal line. **(C)** Statistically, the average SNR of the SSVEP signal was considerably improved after NFT in the experimental group. The SNRs of the first test in the training group and both tests in the control group were similar, while the second test of the experimental group revealed a significant improvement in the SNR. This is a phenomenon not observed in the control group, excluding the possibility that improved skill in the use of the SSVEP-based BCI system is the main reason for the improvement in the experimental group. **(D)** The figure shows the SNR changes in each frequency band in the experimental group and the control group. The experimental group improved in every frequency band, while the changes in the control group appeared to be random. **(E)** Topographic maps of SSVEP frequency responses and P-values from paired-sample *t*-tests. The color change of the posttest map for the experimental group indicates that the training achieved the purpose of reducing interference. For the experimental group, differences ($p < 0.1$) were observed in the following locations: AF3, FC4, CP1, Pz, O2. Importantly, O2 was significantly different ($p < 0.05$).

lobes. Furthermore, the changes in the occipital region implied that the interference there was suppressed and that the SNR increased. The change in the parietal lobe may have been related to the enhancement of the executive control network via neurofeedback regulation. As for the frontal lobe, the difference may have been due to the increased concentration of the subjects' attention and the decrease in disruptive interference.

Network analysis results

From the statistical map of the frequency response difference between the two SSVEP tests, significant differences can be seen in the right parietal lobe, occipital region and frontal lobe; accordingly, AF3, FC4, CP1, and O2 were selected as the four regions of interest to analyze the cortex (result shown in Fig. 9). Overall, in the connected network of the SSVEP task, the parietal lobe is a hub of information transformation, and there is a large amount of information flowing from

the parietal lobe to the occipital lobe and the frontal lobe. This result is also consistent with previous studies (Yan and Gao, 2011).

After calculation, the pre- and posttest information for each node in the test group was compiled as shown in Tables 2 and 3; the rows of the tables represent the origins of information flow, and the columns represent the destinations. Table 4 suggests that CP1, a node in the parietal lobe, contributes the most to network information transmission. According to Tables 2 and 3, the training strengthened the connection of this parietal lobe node to O2 and FC4.

As the information flowed from the parietal lobe to the occipital lobe, we further computed the alpha power of the occipital lobe. As shown in the above figures, the occipital lobe alpha power in the task state was significantly reduced after the use of NFT to upregulate parietal lobe alpha oscillations, while the control group did not significantly change. The decrease in background alpha power in the occipital lobe reduced the frequency-

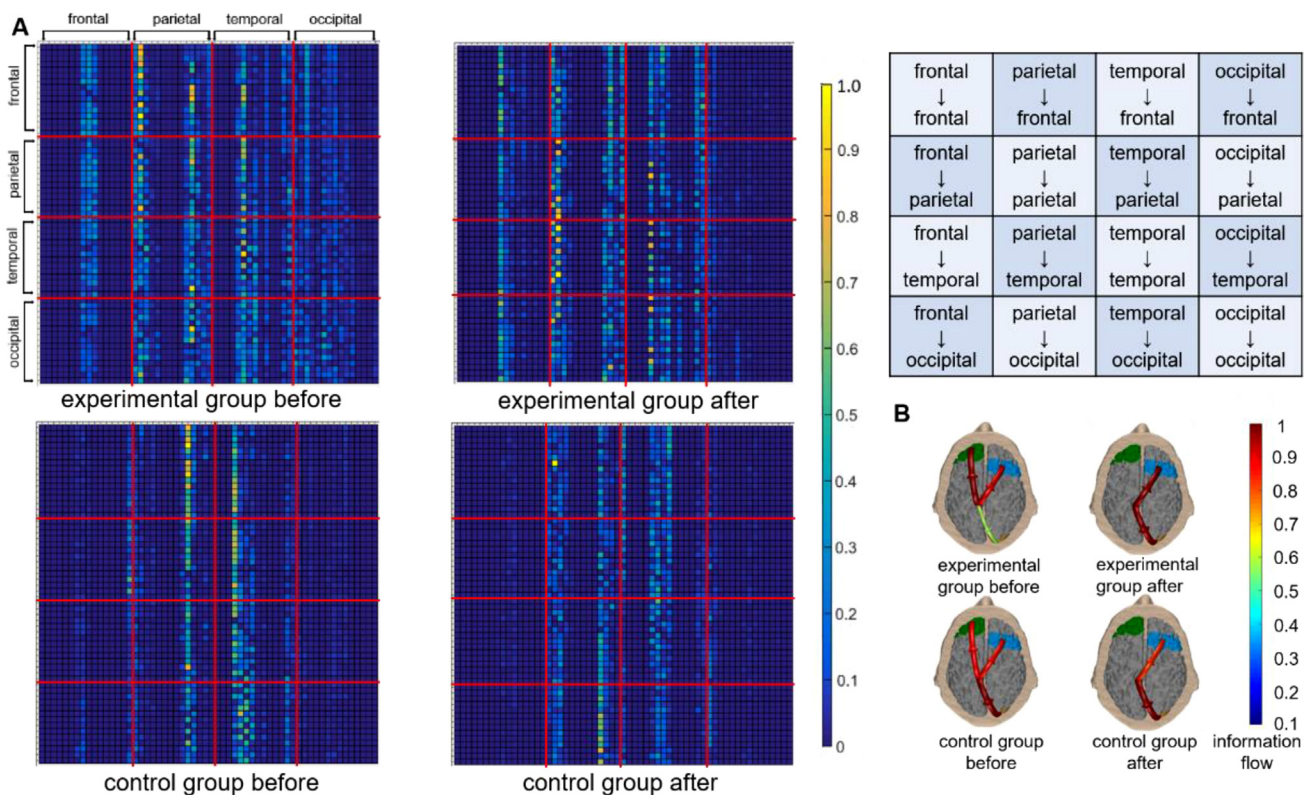


Fig. 9. Results regarding information flow. **(A)** The images illustrate the detailed brain information flow of the two groups of subjects in the SSVEP-based task state. The elements of each matrix are 59×59 electrode leads, and the subareas have been marked. The rows of the graphs represent the origins of information flow, and the columns represent the destinations. Almost all the main hubs in the network transmit information to the whole brain, while the unimportant nodes have little information exchange. **(B)** The figure shows the information flow of cortical functional connectivity in the SSVEP task among the right parietal lobe, occipital region, and frontal lobe. The information flow was calculated using the directed transfer function (DTF).

Table 2. The node information flow of the experimental group (first test/second test)

	AF3	FC4	CP1	O2
AF3	0/0	0/0.01	0.88/0.15	0.01/0
FC4	0.03/0.03	0/0	0.81/0.9	0.01/0
CP1	0.01/0.01	0/0	0/0	0/0
O2	0.04/0.07	0/0.01	0.47/0.9	0/0

Table 3. The node information flow of the control group (first test/second test)

	AF3	FC4	CP1	O2
AF3	0/0	0/0.01	0.81/0.34	0/0
FC4	0.19/0.2	0/0	0.8/0.65	0.01/0
CP1	0.09/0.03	0/0	0/0	0.02/0
O2	0/0.33	0/0	0.84/0.77	0/0

domain correlation between signal and background noise, which was also consistent with our SNR analysis results.

DISCUSSION

As hypothesized, NFT succeeded in strengthening the regulation of parietal lobe activity. The results

demonstrated that the training method in this study was reinforced the executive control network and thus enhanced the ability to complete the task. At the same time, the designed test scale also verified the effectiveness of the training in improving attention.

The alpha band provides active inhibition (Mathewson et al., 2009; Jensen and Mazaheri, 2010; Klimesch, 2012); conversely, downregulation of the alpha band reflects a release from inhibition. The underlying principle of the SSVEP paradigm is that a stimulus target flickering at a fixed frequency induces an evoked response with stable frequency characteristics in the occipital lobe. Enhancement of alpha-band activity in the parietal lobe inhibits task-irrelevant information, thus directing attention (as shown in Table 1), increasing the SNR, and improving SSVEP performance. However, there was no clear correspondence found among the increases in the alpha-band power, SNR, and ITR in the data analysis. This finding indicated that there might be some unknown complex factors in the attention network.

Comparison of similar work and neural mechanism

There have been several previous studies similar to ours (Hanslmayr et al., 2005a; Hanslmayr et al., 2005b; Vernon, 2005; Zoefel et al., 2011; Lee et al., 2019). Benedikt reported that upregulating the amplitude of the upper

Table 4. The increase in information flow by node

	experimental group (first test/ second test)	control group (first test/second test)
AF3	0.08/0.84	0.34/1.49
FC4	0/0.02	0/0.04
CP1	329.63/145.27	22.87/68.64
O2	0.03/0	0.04/0

alpha band in the parieto-occipital lobe successfully enhanced cognitive ability. Benedikt attributed this success to the close relationship between the alpha band and cognition but did not offer an explanation of how the alpha band was associated with cognition. Another work was based on NFT to inhibit alpha interference in order to improve SSVEP performance (Wan et al., 2016). A study by Wan produced results that seem to conflict with the our current findings; it is worthwhile to discuss the difference between the two works in detail. In Wan's study, a two-day NFT regimen was conducted in 20 subjects to inhibit alpha interference in the occipital region, and it was found that this training affected SSVEP performance. One of the distinctions between our experiment and Wan's is that different lobes were regulated. The present study focuses on the parietal lobe, while Wan's research focused on the occipital lobe; however, despite focusing on different aspects, both studies pursued the same goal of enhancing performance with an SSVEP-based BCI.

The present study did not aim to improve BCI operation directly by improving the signal quality; instead, the goal was to improve the subjects' cognitive control. Therefore, the training and the SSVEP test are two independent of the experiment. A frequency band of 8–15 Hz was selected for the SSVEP-based BCI because sensitivity is highest in this interval.

In the SSVEP test conducted by Wan, 7.05, 7.5, 8, 8.57, 9.23, 10, 10.9, 12, 13.33, and 15 Hz were chosen as the stimulation frequencies. The reason for these selections is that the SSVEP response was most active in the low-frequency band (8–16 Hz). Since the frequency range selected for this method includes 8–12 Hz, the background alpha-band activity could be regarded as natural interference. The principle behind the implementation of SSVEP-based BCIs relies on detecting the occipital response matching the stimulation frequency to identify the target. Therefore, the accuracy of a BCI depends heavily on whether the frequency response is precise. The alpha band is usually defined as 8–12 Hz. By the concept of the individual alpha band, this band can extend from roughly 7–13 Hz. The alpha-band component of spontaneous background EEG can interfere with the identification and judgment of SSVEPs in an SSVEP-based BCI. Therefore, Wan's study aimed to downregulate the alpha component of the occipital region through NFT to weaken the interference, which achieved its purpose of improving the operation of an SSVEP-based BCI. More interestingly, analysis of our data showed that the background (task-state) alpha component of the occipital lobe also decreased after

being modulated in the parietal lobe (result shown in Fig. 10). Consistent with our results, Wan decreased the alpha amplitude in the occipital lobe to reduce interference, increase the SNR, and improve performance. It is also worth noting that the method of Wan improved the SNR specifically in the frequency band near the alpha band (8–12 Hz). In our work, it was found that improving the alpha amplitude of the parietal lobe improved the SNR in every frequency band (8–16 Hz). Differences were not limited to a specific frequency band; this finding further confirmed the macro-scale control function of the parietal lobe.

In previous research on SSVEP in our laboratory, it was observed that the SSVEP effect was strictly related to the subjects' attention. This finding is well aligned with the common-sense notion that people are more efficient and effective when they are focused on their tasks. The alpha band is closely linked to attention because it plays a role in suppressing task-related noise. Therefore, increasing the alpha amplitude of the parietal lobe improves attention and enhances the quality of the SSVEP signal. It is worth noting that although two studies have taken the opposite regulatory approaches (upregulation and downregulation), NFT has been applied to different brain lobes and has achieved the same purpose. Based on past reports with numerous observations (Matsumoto and Tanaka, 2004; Lakatos et al., 2005), neuronal systems in the brain can be divided into control networks and execution networks, in which oscillations at various frequencies play different roles. That is, low-frequency oscillations (theta/alpha) are responsible for monitoring, regulation and information transmission, while high-frequency oscillations (beta) subserve information processing and function execution. From this point of view, Wan's protocol and ours may regulate the functional network and control network, respec-

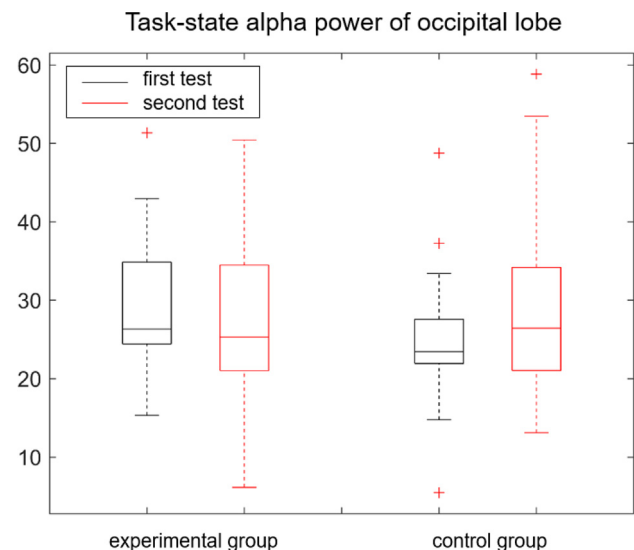


Fig. 10. The picture shows the background alpha component of the occipital lobe (in the task state). As shown in figure, the alpha power in the occipital lobe of the experimental group decreased significantly, and this change was significant according to a paired-sample *t*-test ($t = 23$, $p = 0.0461$).

tively. The human brain does not merely give passive responses to stimuli; it employs a top-down processing mechanism with macro-scale control. In this mechanism, information flows from the top down and distinguishes between control and function, and the overall control also determines the functions of the different regions (Nunez, 2000; Engel et al., 2001). Some research supports the view that the alpha band is involved in the process by which the human brain converts input visual and auditory information into behavior, from control to functional differentiation (Pineda, 2005; Thut et al., 2006; Sauseng et al., 2006).

Further description of the control and functional networks

Prior research has provided extensive direct or indirect evidence that the control network and functional network may exist. Foxe and Bauer support models that implicate parieto-occipital areas prominently in the direction and maintenance of visual attention (Foxe et al., 1998; Bauer, 2006). Meanwhile, evidence has indicated that the control and functional networks might not be wholly separated. In the corresponding difference analysis of SSVEP data before and after training (see Fig. 10), it can be observed that NFT affects the parietal lobe, occipital lobe, and frontal lobe. The alterations that take place in the occipital lobe and the frontal lobe are due to the increase in attention and the decrease in signal noise. Meanwhile, the differences that take place in the parietal lobe suggest that NFT enhances the capacity of the executive control network. In Srinivasan's fMRI studies (Srinivasan et al., 2007), it was also found that some of the occipital voxels were positively correlated with the frontal voxels in a large-scale functional network when visual input was delivered. Owing to these observations and the immense influence of the parietal lobe in regulating the response to visual stimuli, we suggest that the human brain responds to visual stimuli by activating a parietal-occipital-frontal control/functional network in which these three lobes work as an inseparable system.

The outcome of our network connection analysis also confirmed the inference that the frontal lobe, parietal lobe and occipital lobe work in a network within the SSVEP visual task. The parietal lobe serves as the sole hub of information output and transmission in this network. In Tables 2 and 3, it is evident that the information output of the parietal lobe (CP1 to O2, CP1 to FC4) is increased after training in the NFT group, which also confirms that this intervention enhances the control capability of the parietal lobe.

Value and innovativeness of the present study

In our work, a total of 24 children received NFT for two months. After consulting the literature, we found that 1–2 months seems to be the shortest period of NFT that can produce a stable effect. Some previous research suggests that resting alpha activity returns to the initial level after a 20- to 40- minute rest from alpha-band NFT (Ros et al., 2013; Kluetsch et al., 2014), such that the effect of short-term NFT is unstable. Although the EEG

recordings used to calculate the alpha-band power ratio were collected ten days after the last training in this experiment, the outcome showed that 22 of the 24 subjects retained various degrees of improvement. This observation implies that the methods of training in this research produce stable, potentially long-term effects, which has not been reported in many prior studies (Heinrich et al., 2007; Gruzelier, 2013; Gruzelier and John, 2014). Two of the branches of neural engineering with the most research potential are the interpretation of brain information and neuromodulation. In particular, the regulation of neural activity is one of the keys to realizing brain-computer interaction in the future. To realize brain-computer interactions and enhance human brain function, stable and sustainable regulation is crucial. Short-term and unstable modulation methods may cause unexpected adverse effects.

Despite extensive interest in research on the human brain, breakthroughs in brain science must be driven by prudent and ethical experiments. This requires a controllable and stable experimental design to avoid any possible damage or risk. The choice of child subjects in this study undoubtedly increases the difficulty of research. Nonetheless, given the superior neuroplasticity of young brains, this effort is sensible. In most studies on NFT, adults are chosen as subjects (Angelakis et al., 2007; Zotev et al., 2014); similar research on children is relatively rare.

In this work, we developed a novel NFT method and obtained the following conclusions. (1) The effect of NFT can remain stable for a long period after the training ends. (2) Neurofeedback training to improve parietal alpha oscillation can improve attention by suppressing interference. (3) The NFT regimen used in this study can improve the SNR and ITR of the SSVEP signal. (4) This training method can improve subjects' performance with an SSVEP-based BCI. (5) The mechanism of the brain may include control and functional networks that can be adjusted separately.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

A written consent form was completed by all participants' guardians. This project was reviewed by the Ethics Committee of the Tsinghua University School of Medicine (Protocol number 2019006 approved on 3/4/2019).

CONSENT FOR PUBLICATION

Consent to publish the results was obtained from the parent or legal guardian of each subject.

AVAILABILITY OF SUPPORTING DATA

The raw/processed data required to reproduce these findings cannot be shared at this time, as the data are also part of an ongoing study. We will consider publishing some of the essential data once the ongoing study is published.

COMPETING INTERESTS

The authors declare that they have no competing interests.

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AUTHORS' CONTRIBUTIONS

J. Sun contributed to the entire research process, including the experimental design, data collection, result analysis and article writing. J. He participated in the data collection and helped to improve and polish the article. As the corresponding author, Prof. X. Gao guided the research and the writing of the article. He also critically revised the manuscript content and verified the quality of the research. All authors read and approved the final manuscript.

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